

Universidad de Costa Rica

Facultad de Ingeniería

Escuela de Ciencias de la Computación e Informática

Bases de Datos Avanzadas

Laboratorio de MongoDB

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1. Descripción de MongoDB

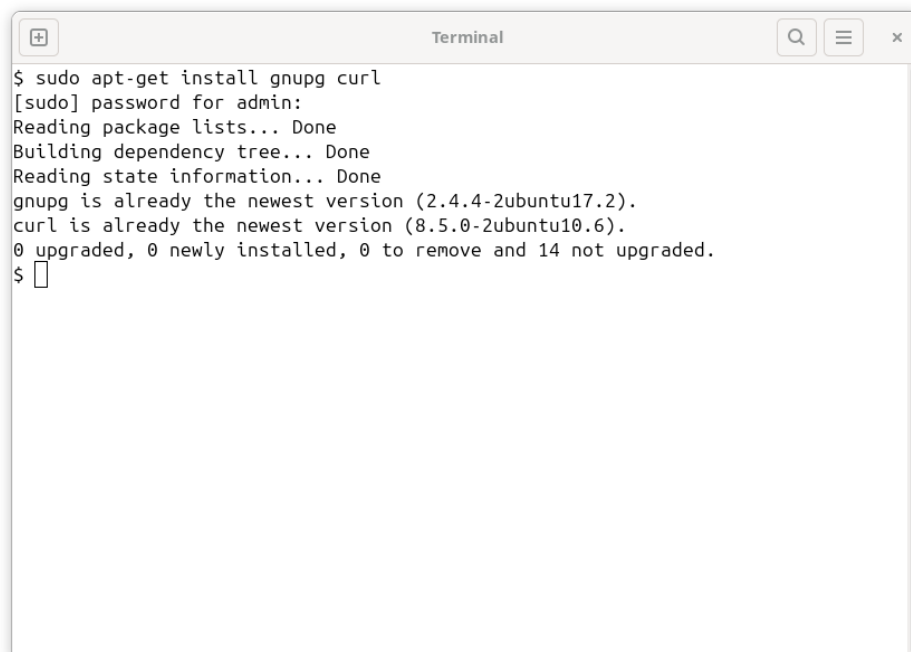
MongoDB es una base de datos NoSQL orientada a documentos que almacena datos en formato BSON, lo que ofrece esquemas flexibles y escalables. Soporta replicación, particionamiento horizontal (*sharding*) y transacciones ACID desde la versión 4.0 [1][2]. Es ampliamente usada en aplicaciones web y microservicios [3].

2. Descripción y ejecución de comandos

A continuación se detalla el propósito de cada comando y se presenta la evidencia de su ejecución.

a. `sudo apt-get install gnupg curl`

Instala `gnupg` (manejo de llaves GPG) y `curl` (cliente para descargas HTTP/S). Ambos son prerequisites para añadir repositorios firmados y trasladar archivos de forma segura [4][5].

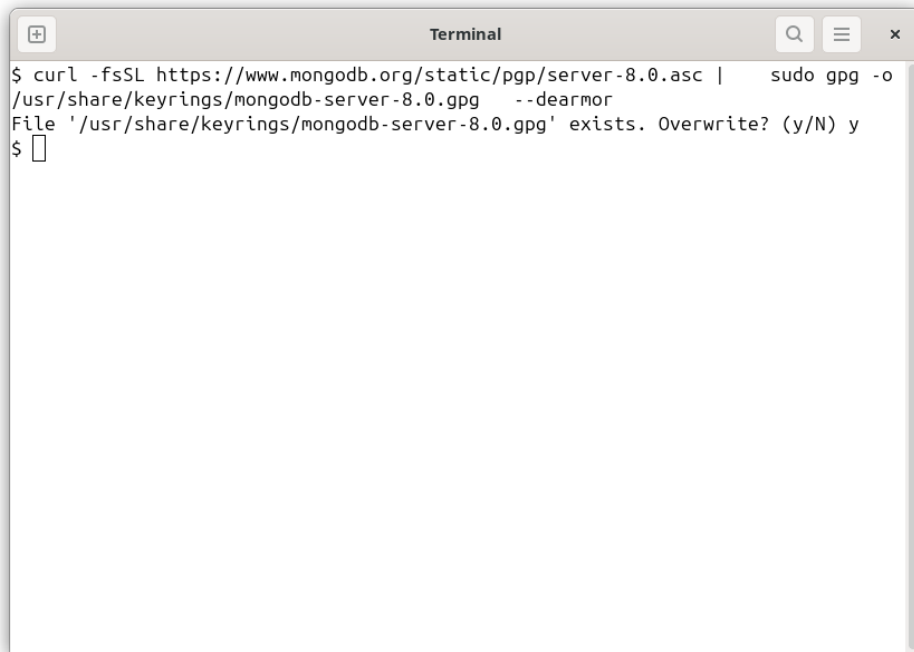


```
$ sudo apt-get install gnupg curl
[sudo] password for admin:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gnupg is already the newest version (2.4.4-2ubuntu17.2).
curl is already the newest version (8.5.0-2ubuntu10.6).
0 upgraded, 0 newly installed, 0 to remove and 14 not upgraded.
$
```

Figura 1: Instalación de `gnupg` y `curl`.

b. `curl -fsSL ... | sudo gpg -o /usr/share/keyrings/mongodb-server-8.0.gpg --dearmor`

Descarga la llave PGP oficial de MongoDB 8.0 (`-fsSL` = silencioso, fallo seguro) y la convierte a formato `.gpg`. Esto garantiza la integridad de los paquetes del repositorio [7].

A terminal window titled "Terminal" with search, menu, and close buttons. It shows the command to download a PGP key from MongoDB's website and convert it to a GPG keyring. The command is: `$ curl -fsSL https://www.mongodb.org/static/pgp/server-8.0.asc | sudo gpg -o /usr/share/keyrings/mongodb-server-8.0.gpg --dearmor`. A prompt message says: `File '/usr/share/keyrings/mongodb-server-8.0.gpg' exists. Overwrite? (y/N) y`. The user has entered 'y' and the prompt is now `$`.

```
$ curl -fsSL https://www.mongodb.org/static/pgp/server-8.0.asc | sudo gpg -o /usr/share/keyrings/mongodb-server-8.0.gpg --dearmor
File '/usr/share/keyrings/mongodb-server-8.0.gpg' exists. Overwrite? (y/N) y
$
```

Figura 2: Descarga y conversión de la llave PGP.

- c. Añade el archivo `mongodb-org-8.0.list` con la línea `deb [...]` que apunta al repositorio oficial para Ubuntu 24.04 (*noble*) y asocia la firma creada en (b) [7].

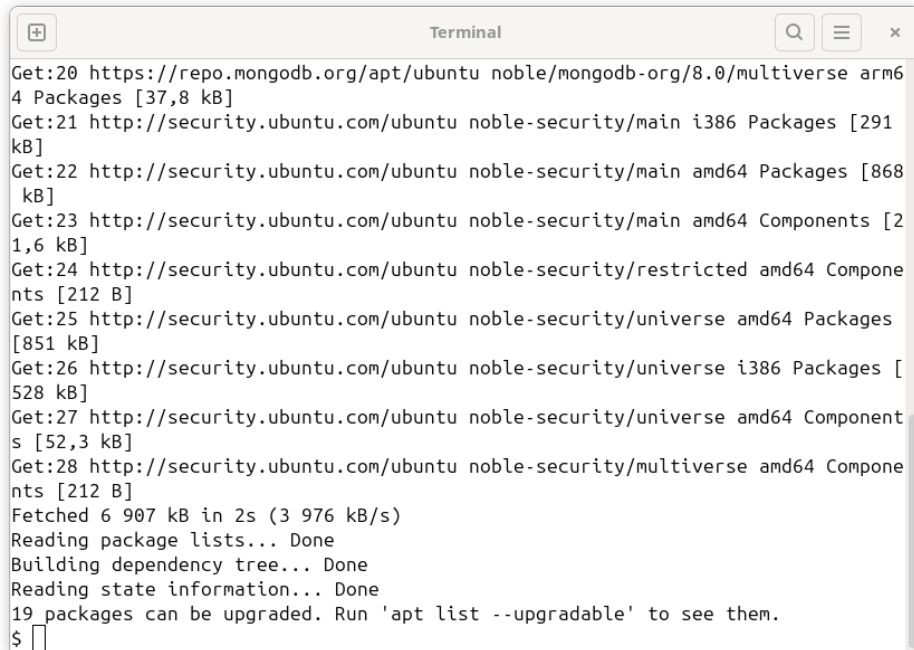
A terminal window titled "Terminal" with search, menu, and close buttons. It shows the command to create a repository file. The command is: `$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-8.0.list`. The output shows the file being created with the specified content. The prompt is now `$`.

```
$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-8.0.list
deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 multiverse
$
```

Figura 3: Creación del archivo de repositorio.

- d. `sudo apt update`

Refresca la base de datos local de paquetes — incluyendo ahora el repositorio de MongoDB — para que el sistema conozca las versiones más recientes [4].

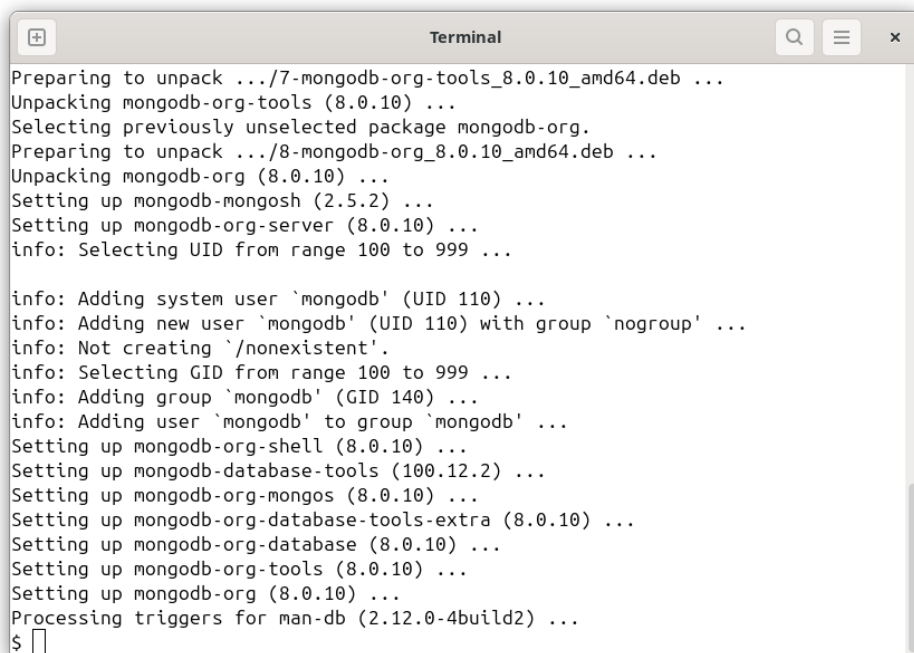


```
Get:20 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse arm64 Packages [37,8 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [291 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [868 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21,6 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [851 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [528 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52,3 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 6 907 kB in 2s (3 976 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
$
```

Figura 4: Actualización de índices de paquetes.

e. `sudo apt-get install -y mongodb-org`

Instala el metapaquete `mongodb-org` (servidor `mongod`, shell, utilidades y archivos de configuración) versión 8.0 [7].



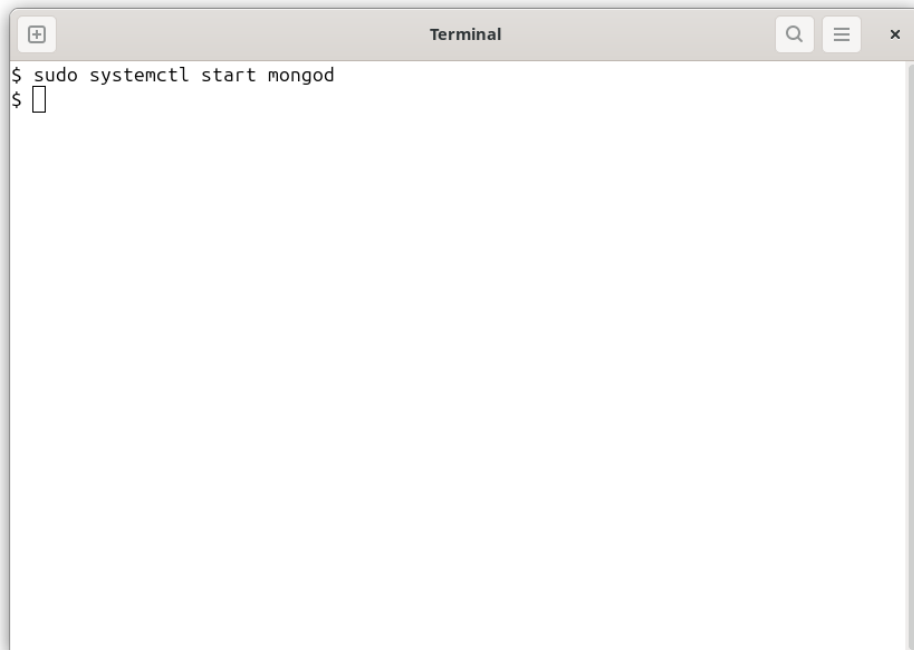
```
Preparing to unpack .../7-mongodb-org-tools_8.0.10_amd64.deb ...
Unpacking mongodb-org-tools (8.0.10) ...
Selecting previously unselected package mongodb-org.
Preparing to unpack .../8-mongodb-org_8.0.10_amd64.deb ...
Unpacking mongodb-org (8.0.10) ...
Setting up mongodb-mongosh (2.5.2) ...
Setting up mongodb-org-server (8.0.10) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `mongodb' (UID 110) ...
info: Adding new user `mongodb' (UID 110) with group `nogroup' ...
info: Not creating `/nonexistent'.
info: Selecting GID from range 100 to 999 ...
info: Adding group `mongodb' (GID 140) ...
info: Adding user `mongodb' to group `mongodb' ...
Setting up mongodb-org-shell (8.0.10) ...
Setting up mongodb-database-tools (100.12.2) ...
Setting up mongodb-org-mongos (8.0.10) ...
Setting up mongodb-org-database-tools-extra (8.0.10) ...
Setting up mongodb-org-database (8.0.10) ...
Setting up mongodb-org-tools (8.0.10) ...
Setting up mongodb-org (8.0.10) ...
Processing triggers for man-db (2.12.0-4build2) ...
$
```

Figura 5: Instalación de `mongodb-org`.

f. `sudo systemctl start mongod`

Inicia el servicio `mongod` mediante `systemd`. Abre el puerto 27017 por defecto [6][8].

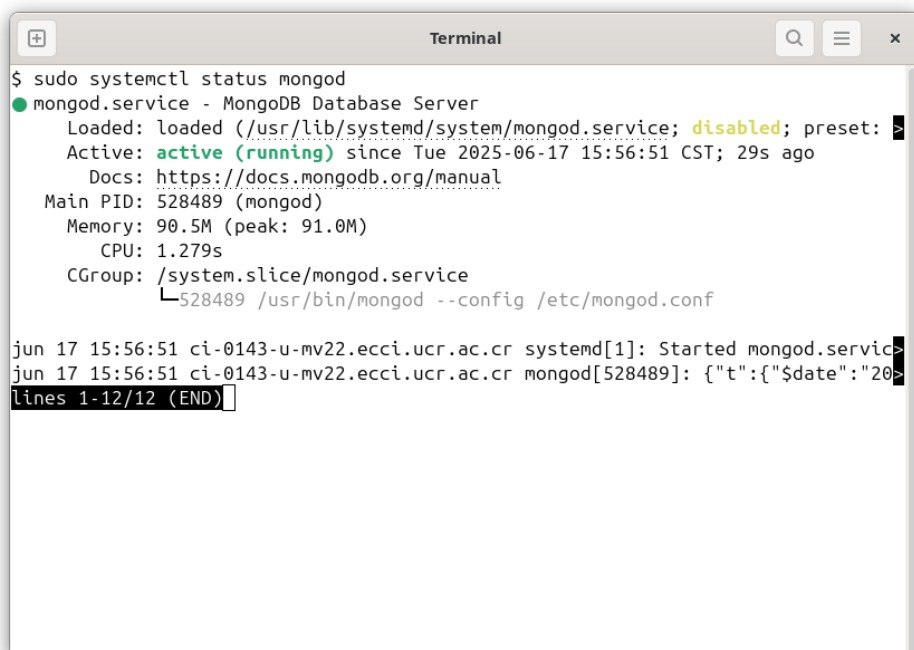


```
Terminal
$ sudo systemctl start mongod
$
```

Figura 6: Inicio del servicio `mongod`.

g. `sudo systemctl status mongod`

Muestra estado, PID, uso de memoria y mensajes recientes del log del servicio [6].



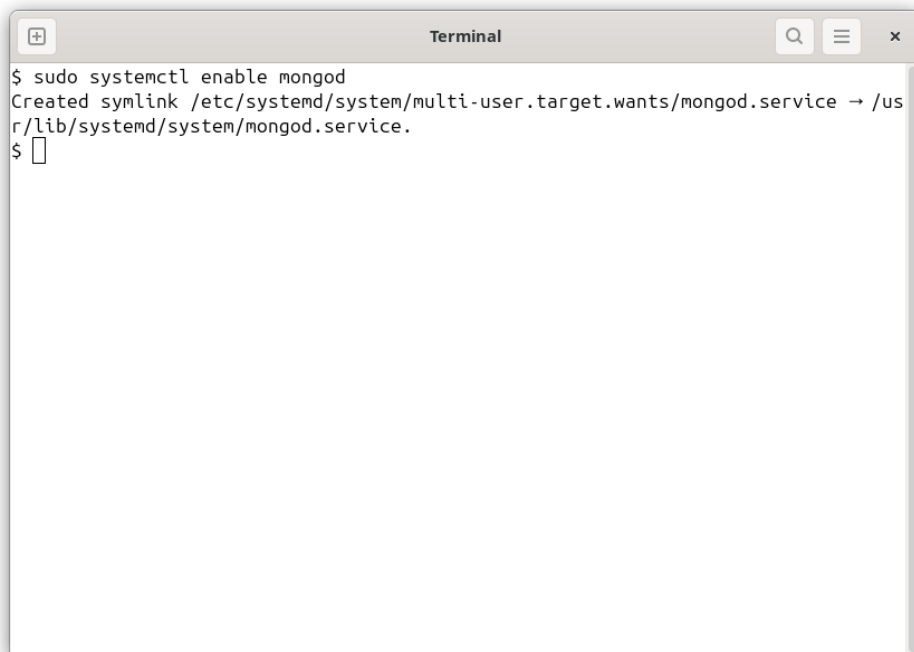
```
Terminal
$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
   Loaded: loaded (/usr/lib/systemd/system/mongod.service; disabled; preset: >
   Active: active (running) since Tue 2025-06-17 15:56:51 CST; 29s ago
     Docs: https://docs.mongodb.org/manual
    Main PID: 528489 (mongod)
      Memory: 90.5M (peak: 91.0M)
         CPU: 1.279s
    CGroup: /system.slice/mongod.service
            └─528489 /usr/bin/mongod --config /etc/mongod.conf

jun 17 15:56:51 ci-0143-u-mv22.ecci.ucr.ac.cr systemd[1]: Started mongod.servic>
jun 17 15:56:51 ci-0143-u-mv22.ecci.ucr.ac.cr mongod[528489]: {"t":{"$date":"20>
lines 1-12/12 (END)
```

Figura 7: Estado del servicio `mongod`.

h. `sudo systemctl enable mongod`

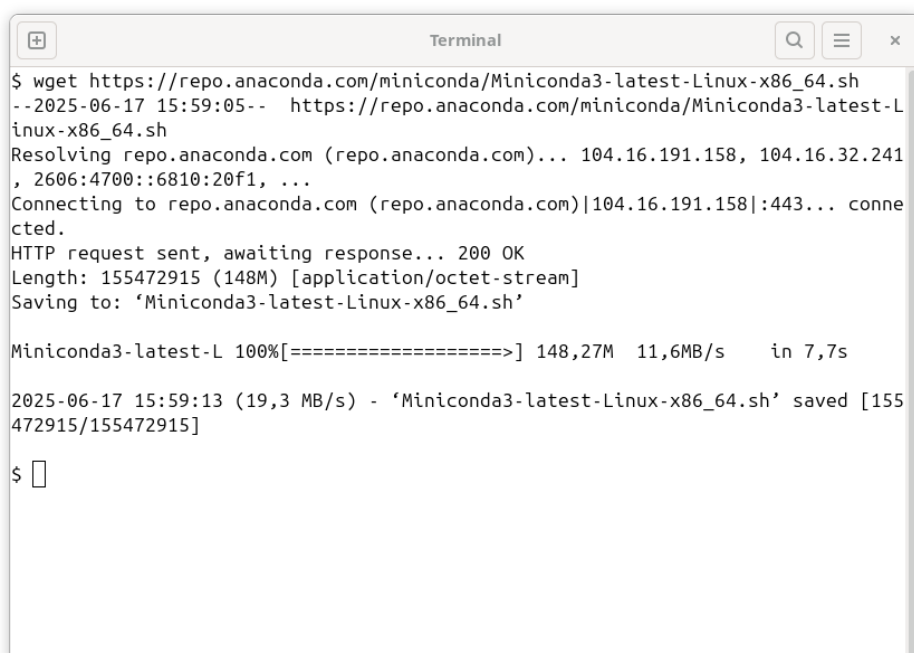
Crea enlaces para que `mongod` se inicie automáticamente al arrancar el sistema [6].



```
Terminal
$ sudo systemctl enable mongod
Created symlink /etc/systemd/system/multi-user.target.wants/mongod.service → /usr/lib/systemd/system/mongod.service.
$
```

Figura 8: Arranque automático de MongoDB.

i. `wget` del script Miniconda (`.sh`). Descarga 150 MB desde los servidores de Anaconda [9].



```
Terminal
$ wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
--2025-06-17 15:59:05-- https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 104.16.191.158, 104.16.32.241, 2606:4700::6810:20f1, ...
Connecting to repo.anaconda.com (repo.anaconda.com)|104.16.191.158|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 155472915 (148M) [application/octet-stream]
Saving to: 'Miniconda3-latest-Linux-x86_64.sh'

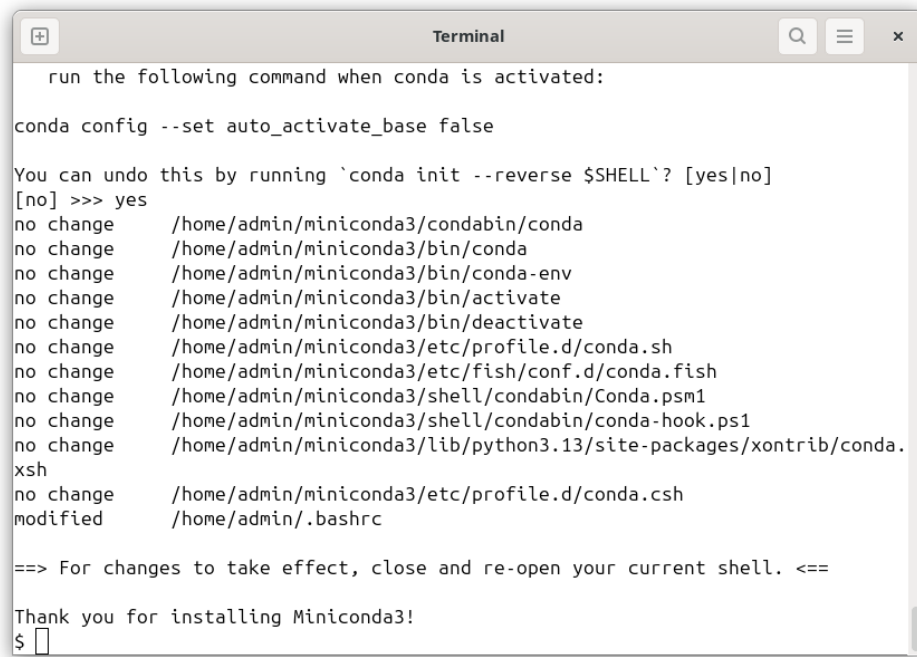
Miniconda3-latest-L 100%[=====>] 148,27M 11,6MB/s in 7,7s

2025-06-17 15:59:13 (19,3 MB/s) - 'Miniconda3-latest-Linux-x86_64.sh' saved [155472915/155472915]

$
```

Figura 9: Descarga de Miniconda.

- j. Ejecución del instalador interactivo de Miniconda; añade `$HOME/miniconda3` y ofrece inicializar conda en `/.bashrc` [9].



```
run the following command when conda is activated:

conda config --set auto_activate_base false

You can undo this by running `conda init --reverse $SHELL`? [yes|no]
[no] >>> yes
no change      /home/admin/miniconda3/condabin/conda
no change      /home/admin/miniconda3/bin/conda
no change      /home/admin/miniconda3/bin/conda-env
no change      /home/admin/miniconda3/bin/activate
no change      /home/admin/miniconda3/bin/deactivate
no change      /home/admin/miniconda3/etc/profile.d/conda.sh
no change      /home/admin/miniconda3/etc/fish/conf.d/conda.fish
no change      /home/admin/miniconda3/shell/condabin/Conda.psm1
no change      /home/admin/miniconda3/shell/condabin/conda-hook.ps1
no change      /home/admin/miniconda3/lib/python3.13/site-packages/xontrib/conda.
xsh
no change      /home/admin/miniconda3/etc/profile.d/conda.csh
modified       /home/admin/.bashrc

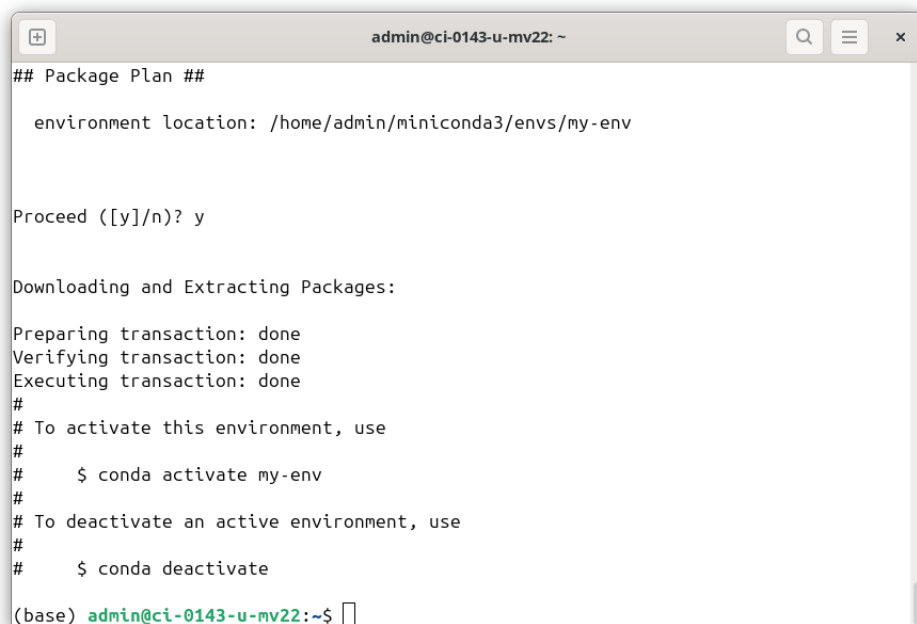
==> For changes to take effect, close and re-open your current shell. <==

Thank you for installing Miniconda3!
$
```

Figura 10: Instalador interactivo de Miniconda.

- k. `conda create -n my-env`

Crea un entorno aislado llamado `my-env` con su propio directorio de paquetes y Python [10].



```
admin@ci-0143-u-mv22: ~
## Package Plan ##

  environment location: /home/admin/miniconda3/envs/my-env

Proceed ([y]/n)? y

Downloading and Extracting Packages:

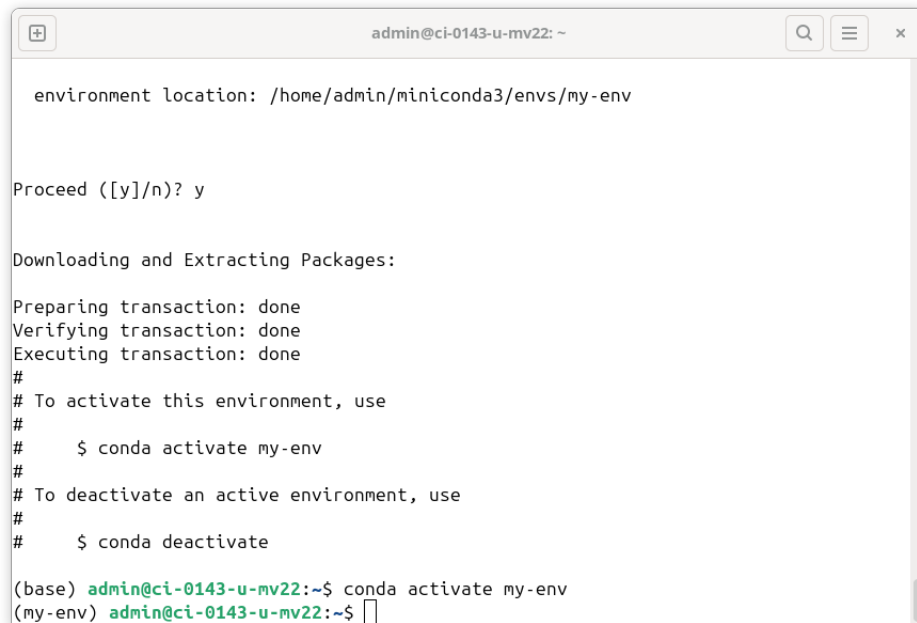
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate my-env
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) admin@ci-0143-u-mv22:~$
```

Figura 11: Creación del entorno `my-env`.

l. conda activate my-env

Ajusta variables de entorno y la ruta para trabajar dentro de my-env [10].

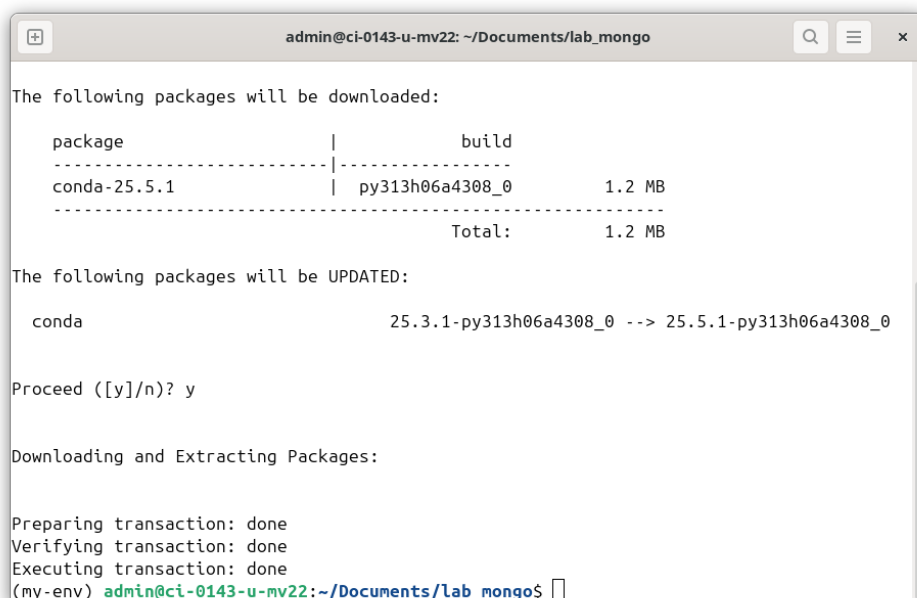


```
admin@ci-0143-u-mv22: ~  
  
environment location: /home/admin/miniconda3/envs/my-env  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
#  
# To activate this environment, use  
#  
#   $ conda activate my-env  
#  
# To deactivate an active environment, use  
#  
#   $ conda deactivate  
  
(base) admin@ci-0143-u-mv22:~$ conda activate my-env  
(my-env) admin@ci-0143-u-mv22:~$
```

Figura 12: Activación del entorno my-env.

m. conda update -n base -c defaults conda

Actualiza el gestor Conda a la versión más reciente en el entorno base; el canal defaults garantiza compatibilidad [10].

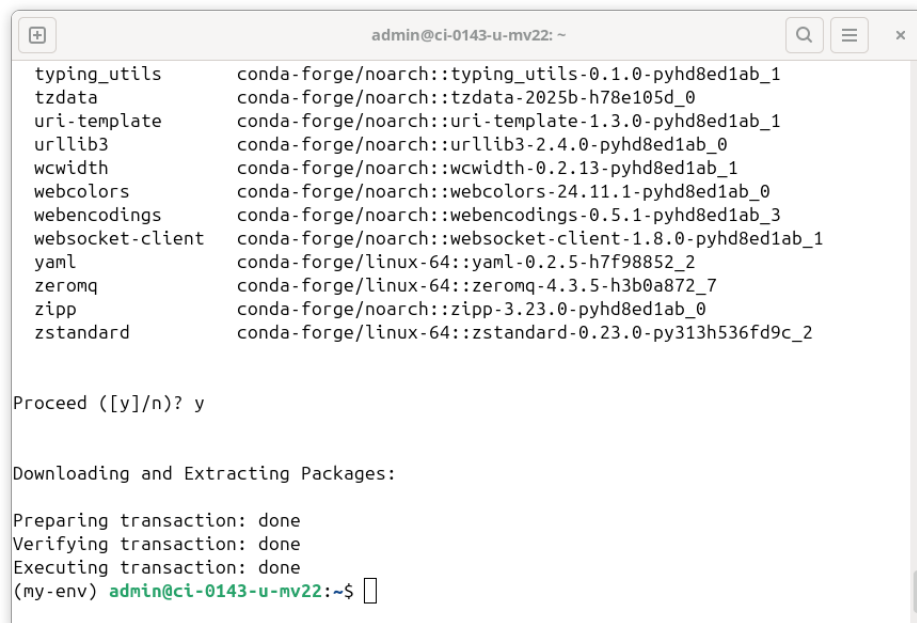


```
admin@ci-0143-u-mv22: ~/Documents/lab_mongo  
  
The following packages will be downloaded:  
  
package | build | size  
-----|-----|-----  
conda-25.5.1 | py313h06a4308_0 | 1.2 MB  
-----|-----|-----  
Total: 1.2 MB  
  
The following packages will be UPDATED:  
  
conda 25.3.1-py313h06a4308_0 -> 25.5.1-py313h06a4308_0  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$
```

Figura 13: Actualización de conda en el entorno base.

n. conda install -n my-env -c conda-forge jupyterlab

Instala JupyterLab dentro de `my-env` desde el canal comunitario `conda-forge` [11].

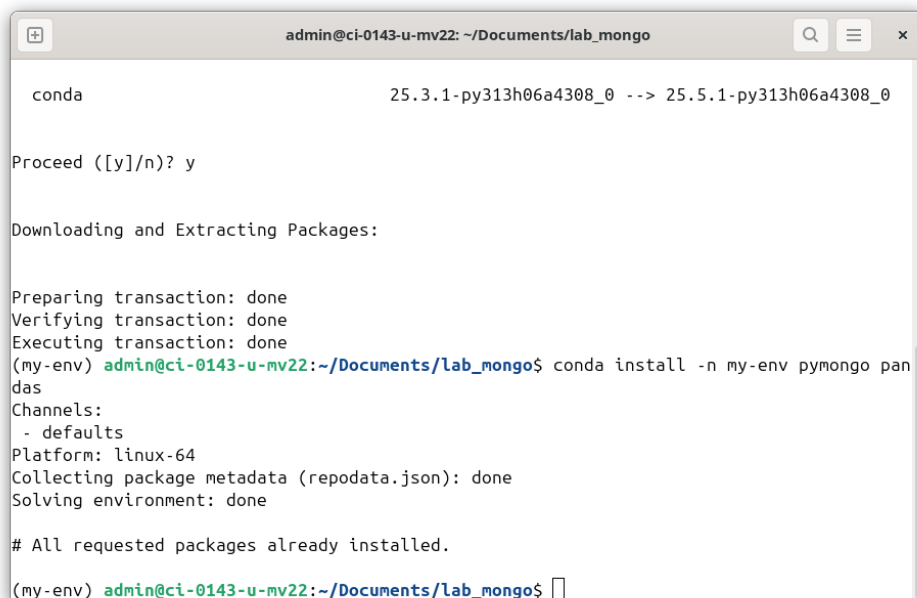


```
admin@ci-0143-u-mv22: ~  
typing_utils      conda-forge/noarch::typing_utils-0.1.0-pyhd8ed1ab_1  
tzdata            conda-forge/noarch::tzdata-2025b-h78e105d_0  
uri-template      conda-forge/noarch::uri-template-1.3.0-pyhd8ed1ab_1  
urllib3           conda-forge/noarch::urllib3-2.4.0-pyhd8ed1ab_0  
wcwidth           conda-forge/noarch::wcwidth-0.2.13-pyhd8ed1ab_1  
webcolors         conda-forge/noarch::webcolors-24.11.1-pyhd8ed1ab_0  
webencodings      conda-forge/noarch::webencodings-0.5.1-pyhd8ed1ab_3  
websocket-client  conda-forge/noarch::websocket-client-1.8.0-pyhd8ed1ab_1  
yaml              conda-forge/linux-64::yaml-0.2.5-h7f98852_2  
zeromq            conda-forge/linux-64::zeromq-4.3.5-h3b0a872_7  
zipp              conda-forge/noarch::zipp-3.23.0-pyhd8ed1ab_0  
zstandard         conda-forge/linux-64::zstandard-0.23.0-py313h536fd9c_2  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my-env) admin@ci-0143-u-mv22:~$
```

Figura 14: Instalación de JupyterLab en `my-env`.

o. `conda install -n my-env pymongo pandas`

Añade las bibliotecas `pymongo` (driver de MongoDB) y `pandas` (análisis de datos) al entorno activo [12][13].

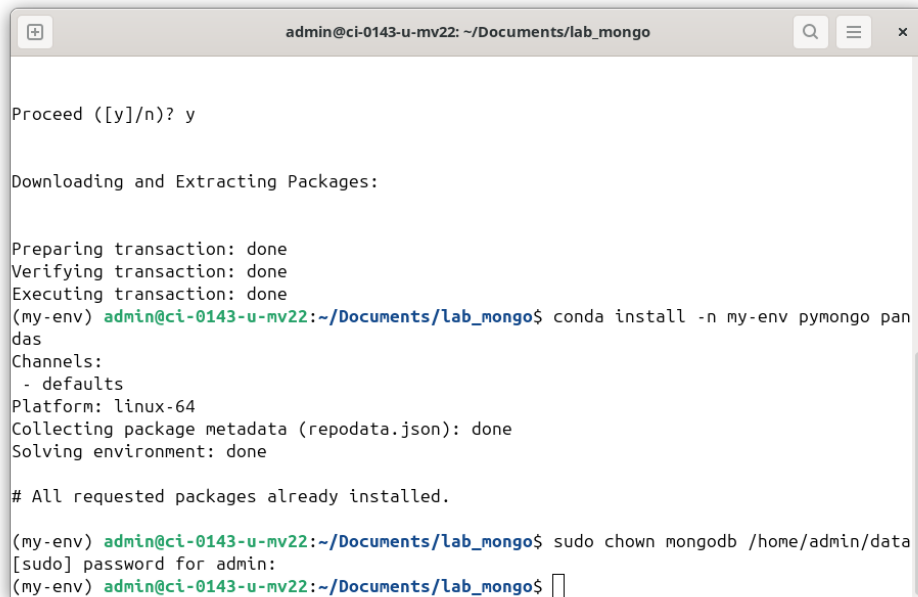


```
admin@ci-0143-u-mv22: ~/Documents/lab_mongo  
conda 25.3.1-py313h06a4308_0 --> 25.5.1-py313h06a4308_0  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ conda install -n my-env pymongo pandas  
Channels:  
- defaults  
Platform: linux-64  
Collecting package metadata (repodata.json): done  
Solving environment: done  
  
# All requested packages already installed.  
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$
```

Figura 15: Instalación de `pymongo` y `pandas`.

p. `sudo chown mongoddb /home/admin/data`

Asigna la propiedad del directorio de datos al usuario del servicio MongoDB, evitando problemas de permisos de lectura/escritura [14].



```
admin@ci-0143-u-mv22: ~/Documents/lab_mongo
Proceed ([y]/n)? y

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ conda install -n my-env pymongo pandas
Channels:
- defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

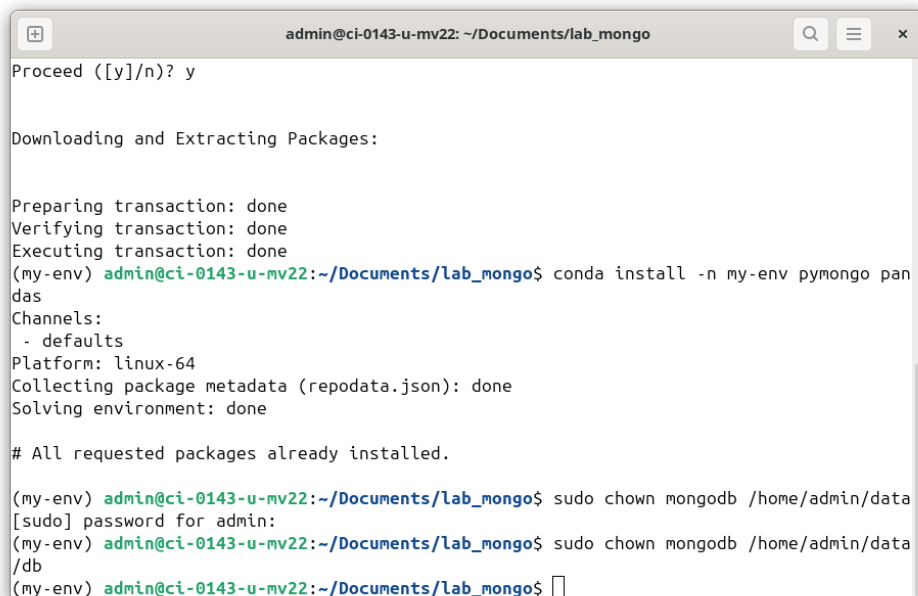
# All requested packages already installed.

(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongod /home/admin/data
[sudo] password for admin:
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$
```

Figura 16: Cambio de propietario de /home/admin/data.

q. `sudo chown mongod /home/admin/data/db`

Asegura que `mongod` pueda escribir directamente en el subdirectorio donde reside la base de datos [14].



```
admin@ci-0143-u-mv22: ~/Documents/lab_mongo
Proceed ([y]/n)? y

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ conda install -n my-env pymongo pandas
Channels:
- defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

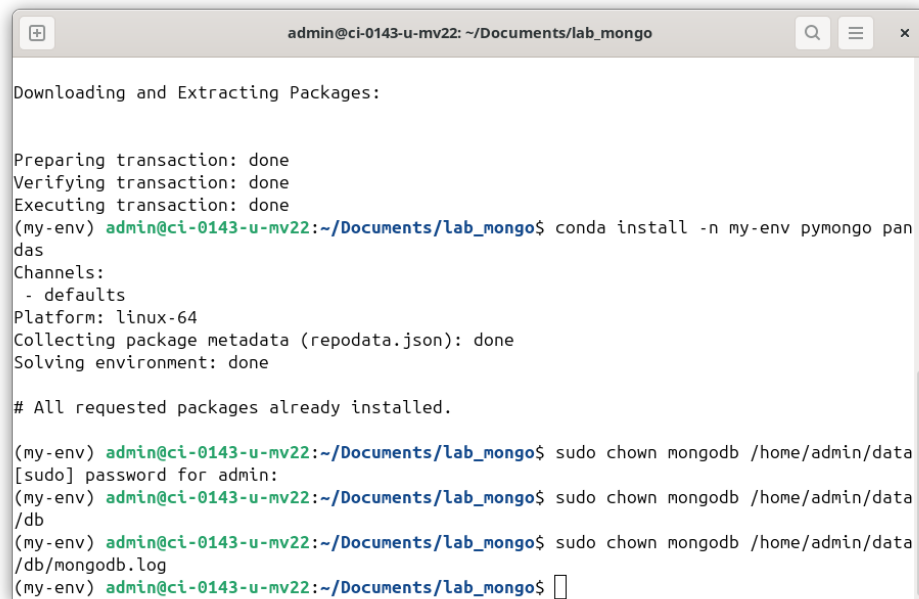
# All requested packages already installed.

(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongod /home/admin/data
[sudo] password for admin:
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongod /home/admin/data/db
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$
```

Figura 17: Propiedad corregida en /home/admin/data/db.

r. `sudo chown mongod /home/admin/data/db/mongodb.log`

Otorga a `mongod` permisos sobre el archivo de log para poder rotarlo y anexar entradas [14].



```
admin@ci-0143-u-mv22: ~/Documents/lab_mongo
Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ conda install -n my-env pymongo pandas
Channels:
- defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

# All requested packages already installed.

(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongodb /home/admin/data
[sudo] password for admin:
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongodb /home/admin/data
/db
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$ sudo chown mongodb /home/admin/data
/db/mongod.log
(my-env) admin@ci-0143-u-mv22:~/Documents/lab_mongo$
```

Figura 18: Permisos de `mongod.log` ajustados.

3. Configurar y ejecutar Jupyter Lab

Los pasos siguientes completan la configuración; las figuras n-r documentan cada fase.

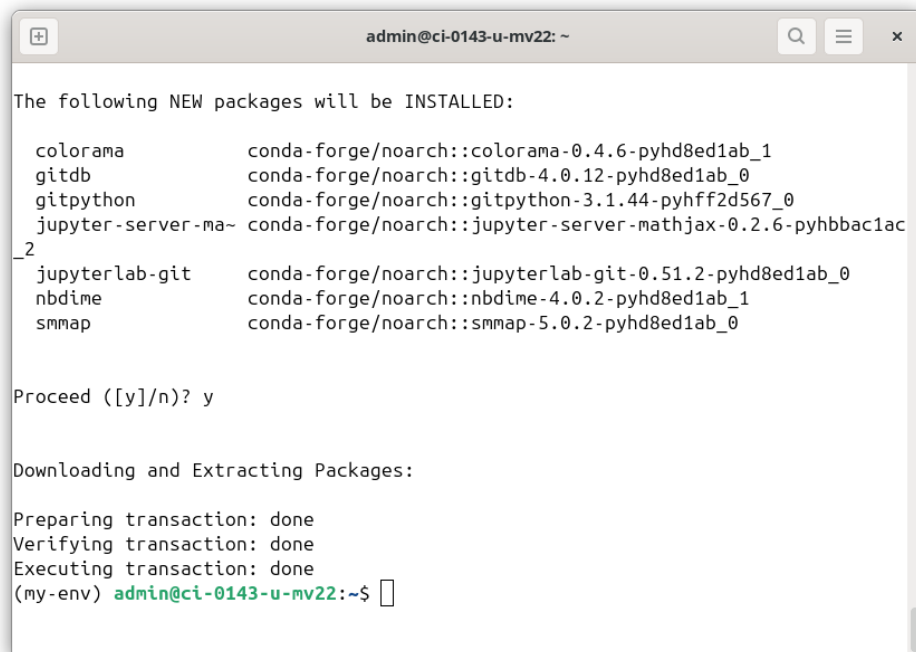
1. Activar el entorno `my-env` (`conda activate my-env`).



```
admin@ci-0143-u-mv22: ~
$ bash
(base) admin@ci-0143-u-mv22:~$ conda activate my-env
(my-env) admin@ci-0143-u-mv22:~$
```

Figura 19: Prompt con `(my-env)` activo.

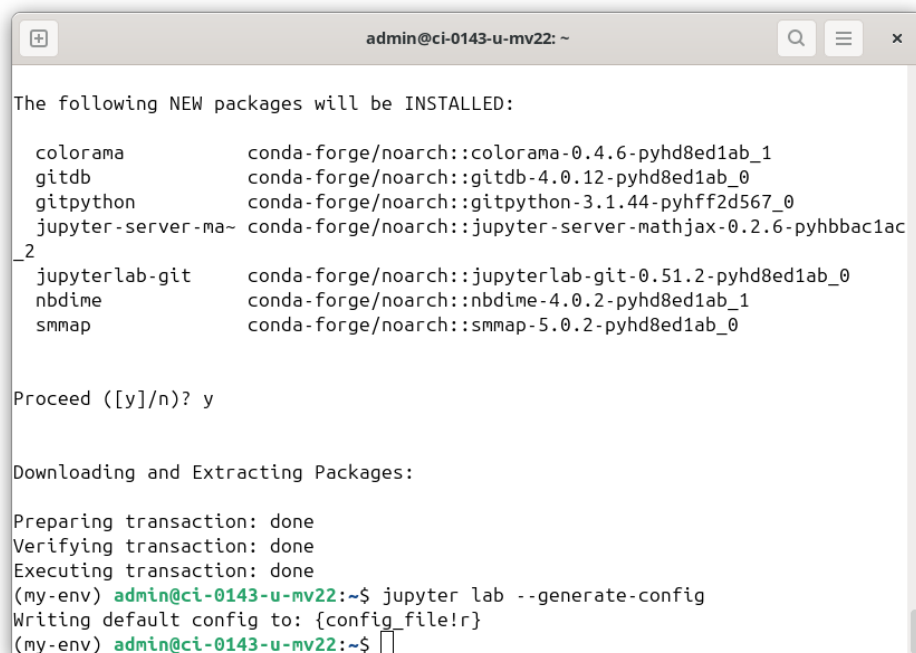
2. **Instalar librerías recomendadas (opcional).** *pandas*, *matplotlib*, *pymongo*, etc.; registrar el kernel e instalar *jupyterlab-git*.



```
admin@ci-0143-u-mv22: ~  
The following NEW packages will be INSTALLED:  
  
colorama      conda-forge/noarch::colorama-0.4.6-pyhd8ed1ab_1  
gitdb         conda-forge/noarch::gitdb-4.0.12-pyhd8ed1ab_0  
gitpython     conda-forge/noarch::gitpython-3.1.44-pyhff2d567_0  
jupyter-server-matplotlib conda-forge/noarch::jupyter-server-matplotlib-0.2.6-pyhbba1ac  
_2  
jupyterlab-git conda-forge/noarch::jupyterlab-git-0.51.2-pyhd8ed1ab_0  
nbdtm         conda-forge/noarch::nbdtm-4.0.2-pyhd8ed1ab_1  
smmap         conda-forge/noarch::smmap-5.0.2-pyhd8ed1ab_0  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my-env) admin@ci-0143-u-mv22:~$
```

Figura 20: Instalación de paquetes adicionales.

3. **Generar archivo de configuración.** `jupyter lab --generate-config`. Ajustar IP, puerto y navegador.



```
admin@ci-0143-u-mv22: ~  
The following NEW packages will be INSTALLED:  
  
colorama      conda-forge/noarch::colorama-0.4.6-pyhd8ed1ab_1  
gitdb         conda-forge/noarch::gitdb-4.0.12-pyhd8ed1ab_0  
gitpython     conda-forge/noarch::gitpython-3.1.44-pyhff2d567_0  
jupyter-server-matplotlib conda-forge/noarch::jupyter-server-matplotlib-0.2.6-pyhbba1ac  
_2  
jupyterlab-git conda-forge/noarch::jupyterlab-git-0.51.2-pyhd8ed1ab_0  
nbdtm         conda-forge/noarch::nbdtm-4.0.2-pyhd8ed1ab_1  
smmap         conda-forge/noarch::smmap-5.0.2-pyhd8ed1ab_0  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages:  
  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my-env) admin@ci-0143-u-mv22:~$ jupyter lab --generate-config  
Writing default config to: {config_file!r}  
(my-env) admin@ci-0143-u-mv22:~$
```

Figura 21: Edición de `jupyter_lab_config.py`.

4. Lanzar Jupyter Lab. La URL /lab?token= abre el “Launcher”.

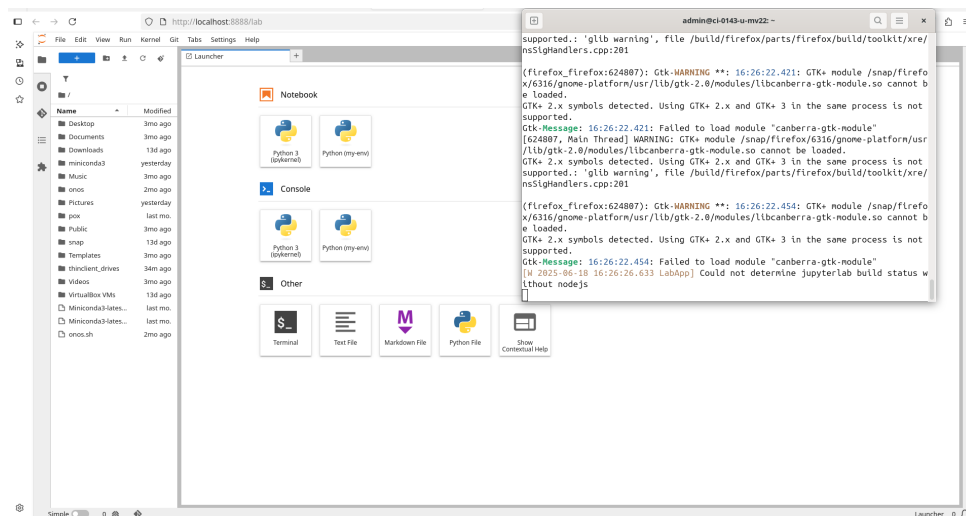


Figura 22: Interfaz de Jupyter Lab.

5. Probar conexión a MongoDB en un notebook.

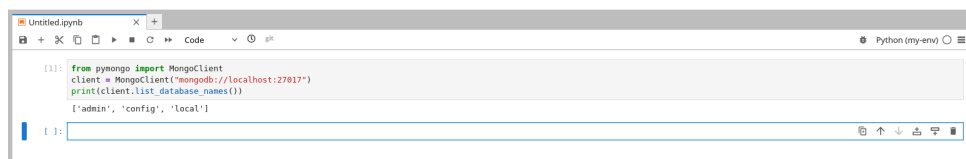


Figura 23: Salida de list_database_names().

6. Cerrar Jupyter Lab (Ctrl + C).

4. Segunda Parte del Laboratorio: Operaciones en Jupyter (Python + PyMongo)

En esta parte se ejecutaron varias celdas de Jupyter para poblar, consultar y manipular la base de datos `sample_db`. Cada celda se documenta con una breve descripción y la captura resultante.

a. Wrapper con control de tiempo (10 s).

Ejecuta tres comandos de sistema (`mkdir`, `mongod`, `mongosh`) con alarma; evita bloqueos si alguno excede el límite. *Resultado:* `mkdir` correctamente, `mongod --fork` devolvió código 1 porque ya existía un demonio activo, y `mongosh` se conectó sin problemas.

```

Running command: mkdir -p /home/admin/data/db
Running command: mongod --dbpath /home/admin/data/db --bind_ip 127.0.0.1 --fork --logpath /home/admin/data/db/mongod.log
about to fork child process, waiting until server is ready for connections.
forked process: 782083
ERROR: child process failed, exited with 1
To see additional information in this output, start without the "--fork" option.
Command failed with error: Command 'mongod --dbpath /home/admin/data/db --bind_ip 127.0.0.1 --fork --logpath /home/admin/data/db/mongod.log' returned non-zero exit status 1.
Running command: mongosh
Current Mongosh Log ID: 68558cd542b027743069e327
Connecting to: mongod://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.2
Using MongoDB: 8.0.10
Using Mongosh: 2.5.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2025-06-17T15:56:52.403-06:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2025-06-17T15:56:53.321-06:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2025-06-17T15:56:53.321-06:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2025-06-17T15:56:53.321-06:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2025-06-17T15:56:53.321-06:00: We suggest setting the contents of sysfsFile to 0.
2025-06-17T15:56:53.321-06:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
test>

```

Figura 24: Salida de la celda a: conexión a mongosh.

b. Inserción de datos de ejemplo.

Crea la BD `sample_db`, colección `products` y carga 10 documentos de inventario.

```

# Sample data to insert into the MongoDB collection
data = [
    {"product_id": 1, "name": "Laptop", "category": "Electronics", "price": 1200, "quantity": 10},
    {"product_id": 2, "name": "Smartphone", "category": "Electronics", "price": 800, "quantity": 25},
    {"product_id": 3, "name": "Tablet", "category": "Electronics", "price": 600, "quantity": 15},
    {"product_id": 4, "name": "Headphones", "category": "Accessories", "price": 150, "quantity": 50},
    {"product_id": 5, "name": "Smartwatch", "category": "Accessories", "price": 300, "quantity": 30},
    {"product_id": 6, "name": "Camera", "category": "Photography", "price": 1000, "quantity": 8},
    {"product_id": 7, "name": "Tripod", "category": "Photography", "price": 100, "quantity": 20},
    {"product_id": 8, "name": "Monitor", "category": "Electronics", "price": 400, "quantity": 12},
    {"product_id": 9, "name": "Keyboard", "category": "Accessories", "price": 50, "quantity": 100},
    {"product_id": 10, "name": "Mouse", "category": "Accessories", "price": 40, "quantity": 75}
]

# Insert sample data into the collection
collection.insert_many(data)
print("Sample data inserted successfully!")

Sample data inserted successfully!

```

Figura 25: Confirmación de inserción en `products`.

c. Consulta simple por categoría.

Filtra los productos con `category = "Electronics"` y los muestra como `DataFrame`.

```
# Function to convert MongoDB query results to DataFrame for display
def to_dataframe(cursor):
    return pd.DataFrame(list(cursor))

# Simple Query - Find all products in Electronics category
electronics_products = collection.find({"category": "Electronics"})
electronics_df = to_dataframe(electronics_products)
electronics_df
```

	_id	product_id	name	category	price	quantity
0	685596fa35d5de5a8c27ea1c	1	Laptop	Electronics	1200	10
1	685596fa35d5de5a8c27ea1d	2	Smartphone	Electronics	800	25
2	685596fa35d5de5a8c27ea1e	3	Tablet	Electronics	600	15
3	685596fa35d5de5a8c27ea23	8	Monitor	Electronics	400	12

Figura 26: Productos de la categoría *Electronics*.

d. Pipeline de agregación para *Accessories*.

\$match + \$sort + \$group: calcula el *total_quantity* y *average_price*.

```
# Aggregation Pipeline Example
pipeline = [
    {"$match": {"category": "Accessories"}},
    {"$sort": {"price": -1}},
    {"$group": {"_id": "$category", "total_quantity": {"$sum": "$quantity"}, "average_price": {"$avg": "$price"}}},
    {"$limit": 2}
]

aggregation_result = collection.aggregate(pipeline)
aggregation_df = to_dataframe(aggregation_result)
aggregation_df
```

	_id	total_quantity	average_price
0	Accessories	255	135.0

Figura 27: Totales y media de la categoría *Accessories*.

e. Actualización de precio (\$set).

Cambia el price del producto #1 (Laptop) de 1200 → 1100 y muestra el documento actualizado.

```
# Update the price of the product with product_id 1 (Laptop)
update_result = collection.update_one({"product_id": 1}, {"$set": {"price": 1100}})
print(f"Matched {update_result.matched_count} document(s) and modified {update_result.modified_count} document(s).")

# Fetch the updated document
updated_product = collection.find_one({"product_id": 1})
updated_product
```

Matched 1 document(s) and modified 1 document(s).

```
{'_id': ObjectId('685596fa35d5de5a8c27ea1c'),
 'product_id': 1,
 'name': 'Laptop',
 'category': 'Electronics',
 'price': 1100,
 'quantity': 10}
```

Figura 28: Laptop con precio actualizado.

f. Eliminación masiva (\$delete_many).

Borra todos los documentos con `category = "Photography"` y lista el inventario restante.

```
: # Delete products in the 'Photography' category
delete_result = collection.delete_many({"category": "Photography"})
print(f"Deleted {delete_result.deleted_count} document(s).")

# Verify remaining documents
remaining_products = collection.find()
to_dataframe(remaining_products)
```

Deleted 2 document(s).

	_id	product_id	name	category	price	quantity
0	685596fa35d5de5a8c27ea1c	1	Laptop	Electronics	1100	10
1	685596fa35d5de5a8c27ea1d	2	Smartphone	Electronics	800	25
2	685596fa35d5de5a8c27ea1e	3	Tablet	Electronics	600	15
3	685596fa35d5de5a8c27ea1f	4	Headphones	Accessories	150	50
4	685596fa35d5de5a8c27ea20	5	Smartwatch	Accessories	300	30
5	685596fa35d5de5a8c27ea23	8	Monitor	Electronics	400	12
6	685596fa35d5de5a8c27ea24	9	Keyboard	Accessories	50	100
7	685596fa35d5de5a8c27ea25	10	Mouse	Accessories	40	75

Figura 29: Inventario después de eliminar *Photography*.

g. Creación de índice en name.

Genera un índice ascendente `name_1` y muestra la lista de índices de la colección.

```
# Create an index on the 'name' field to optimize search queries
index_result = collection.create_index("name")
print(f"Index created: {index_result}")

# List all indexes
indexes = collection.index_information()
indexes
```

Index created: name_1

```
{'_id': {'v': 2, 'key': [['_id', 1]]},
 'name_1': {'v': 2, 'key': [['name', 1]]}}
```

Figura 30: Índices de la colección `products`.

i. Join products ↔ orders con \$lookup.

Crea la colección `orders`, inserta tres pedidos y muestra solo los productos con órdenes asociadas.


```

: # Add a text index on the 'name' field for text search
collection.create_index([("name", "text")])

# Use text search to find products containing the word 'Smart'
text_search_result = collection.find({"$text": {"$search": "Smart"}})
to_dataframe(text_search_result)

```

: —

Figura 31: Resultado del \$lookup entre productos y órdenes.

j. \$project – vista reducida.

Devuelve únicamente los campos `name` y `price` (sin `_id`) para todos los productos.

	_id	product_id	name	category	price	quantity	order_info
0	685596fa35d5de5a8c27ea1c	1	Laptop	Electronics	1100	10	{'_id': 685765af35d5de5a8c27ea26, 'order_id':...
1	685596fa35d5de5a8c27ea1d	2	Smartphone	Electronics	800	25	{'_id': 685765af35d5de5a8c27ea27, 'order_id':...
2	685596fa35d5de5a8c27ea20	5	Smartwatch	Accessories	300	30	{'_id': 685765af35d5de5a8c27ea28, 'order_id':...

Figura 32: Listado simplificado (nombre + precio).

k. \$lookup + \$unwind.

Une `products` con `orders` y aplanar `order_info`, obteniendo una fila por (producto, orden).

```

: # Use $project to include only 'name' and 'price' fields in the output
project_result = collection.aggregate([
    {"$project": {"_id": 0, "name": 1, "price": 1}}
])
to_dataframe(project_result)

```

```

:

```

	name	price
0	Laptop	1100
1	Smartphone	800
2	Tablet	600
3	Headphones	150
4	Smartwatch	300
5	Monitor	400
6	Keyboard	50
7	Mouse	40

Figura 33: Tabla aplanada producto–orden.

l. Búsqueda con expresión regular.

Encuentra productos cuyo `name` comienza con «S» y los muestra como `DataFrame`.

```

: # Use $unwind to flatten the 'order_info' array
  unwind_result = collection.aggregate([
    {
      "$lookup": {
        "from": "orders",
        "localField": "product_id",
        "foreignField": "product_id",
        "as": "order_info"
      },
    },
    {"$unwind": "$order_info"}
  ])
to_dataframe(unwind_result)

```

	_id	product_id	name	category	price	quantity	order_info
0	685769548b29a71cfc1dc0f5	1	Laptop	Electronics	1100	10	{'_id': 68576a4f8b29a71cfc1dc0ff, 'order_id': ...
1	685769548b29a71cfc1dc0f6	2	Smartphone	Electronics	800	25	{'_id': 68576a4f8b29a71cfc1dc100, 'order_id': ...
2	685769548b29a71cfc1dc0f9	5	Smartwatch	Accessories	300	30	{'_id': 68576a4f8b29a71cfc1dc101, 'order_id': ...

Figura 34: Productos cuyo nombre inicia con “S”.

5. Referencias

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